

Medical abbreviation bake-off

CS496 Project 0 | DRAFT - real benchmark not run

1. Task and data

Expand one abbreviation in a short UK medical-record context. Example: "The nurse recorded BP before the consultation."
Expected: blood pressure. Choose one exact string from a fixed 60-term output vocabulary. Correct answers are stored separately and never supplied with questions.

50 test items and 10 development items, with disjoint abbreviations. Factual expansions: NHS glossary; contexts: original synthetic sentences drafted with AI assistance. Parenthetical explanations are omitted and canonical case is fixed. Two people must independently check every label in data/label_reviews.csv; review is pending in the supplied draft. No patient records are used.

2. Experiment

top_api: gemini-3.1-pro-preview

cheap_api: gemini-3.5-flash-lite

local: qwen2.5:1.5b

Same literal prompt, vocabulary, question order, parser, temperature 0 and output limit 2048 for all models. One call per item; sequential requests; no retries, retrieval, tools or chat history. Native tokenisers and internal reasoning defaults differ. Blank, non-vocabulary, extra-text or multiline outputs fail parsing; outer whitespace alone is ignored. Refusals, timeouts, truncation and API failures count as wrong.

Wall-clock latency measures send to full response (all attempted calls, including errors and the first local cold load). p50/p95 use linear interpolation. Local generation speed uses total generated tokens / total generation time; it excludes prompt processing. Exact versions, UTC date and local model digest are saved in run metadata.

3. Results

Model	Correct	p50 / p95 ms	USD / 1k	P / R / T / O
gemini-3.1-pro-preview	pending	pending	pending	pending
gemini-3.5-flash-lite	pending	pending	pending	pending
qwen2.5:1.5b	pending	pending	pending	pending

P=parse errors; R=refusals; T=timeouts; O=other execution errors. All remain in the accuracy denominator. Missing usage means unknown cost, never zero. Source: results/per_item.csv; no model measurements are included in the supplied draft.

Evidence, cost and decision

4. Wrong answers (first three per model)

gemini-3.1-pro-preview — pending real run

gemini-3.5-flash-lite — pending real run

qwen2.5:1.5b — pending real run

5. Cost, traffic and hardware

API cost = (uncached input × input rate + cached input × cache rate + visible and thinking output × output rate) / 1,000,000. Standard paid USD list prices, checked 2026-09-11. Local cost = hardware USD/hour ÷ measured requests/hour plus monthly operator labour divided by volume. Hardware hourly rate must include depreciation/rental and electricity; assumptions remain user-supplied.

Today: 1,000 requests/month; 100x: 100,000. Numeric totals and break-even are pending measurements and real hardware/labour assumptions. Break-even $V = \text{fixed monthly labour} / (\text{API cost/request} - \text{local variable cost/request})$, only when the denominator is positive. Validate against one-machine capacity before interpreting it.

Hardware: pending your PC specifications, RAM, GPU/VRAM and Ollama version. Default local model: Qwen2.5 1.5B, Q4_K_M (record actual digest and quantisation during the run).

6. Choice and limitations

Decision pending real results. Predeclared rule: choose the lowest-cost model reaching at least 95% exact-match accuracy and p95 below 10 seconds; if none qualifies, report that none meets the target. Change the choice if accuracy falls below that threshold, latency exceeds the target, or measured traffic crosses a feasible local/API break-even point. Replace this paragraph with the actual winning model, supporting numbers and trade-offs after the run; do not claim an unmeasured winner.

A vocabulary-constrained benchmark measures canonical matching, not clinical competence. Contexts may make the task easy. If development results saturate, revise the task before freezing; never remove failed test items. With only 50 test cases, one item changes accuracy by two percentage points. Exact-match failures can reflect formatting rather than knowledge.

Sources: NHS, nhs.uk/nhs-app/help/understanding-abbreviations/; Google, ai.google.dev/gemini-api/docs/pricing and [/models](https://ai.google.dev/gemini-api/docs/models); Ollama, ollama.com/library/qwen2.5:1.5b. Full URLs and methodology: [data/labelling_note.md](#) and [README.md](#).